

# D KARTHIK REDDY

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## PROFESSIONAL SUMMARY

A trained engineering junior who worked on Python and C++, design and contribution to the creation of constructive AI, and scalable embedded systems. Studies in deepfake detection, acoustic surveillance, and sensing with the use of drones, because of an interest in end-to-end software design. National visibility in the field of innovation due to published patent and the first place at hackathons. Looking forward to using the latest development and analytics skills to solve a complicated financial technology problem..

## EDUCATION

**VNR Vignana Jyothi Institute of Engineering and Technology**

**2022 – 2026**

*B.Tech in Electronics and Communication Engineering • CGPA: 8.81/10.0*

*JNTUH*

**Minor Degree in Artificial Intelligence and Machine Learning**

**2024 – Present**

*VNR VJIT*

*JNTUH*

**Sri Chaitanya Junior College**

**2020 – 2022**

*Intermediate • Percentage: 98%*

*Board of Intermediate Education*

**Sri Chaitanya School**

**2020**

*SSC • CGPA: 10.0/10.0*

*Board of Secondary Education*

## TECHNICAL SKILLS

- Programming Languages: C++, Python, Java , JavaScript,SQL
- Web Development: HTML, CSS, Bootstrap, React
- Machine Learning: TensorFlow, PyTorch, OpenCV, Scikit-learn
- Embedded Systems: Arduino, Raspberry Pi, ARM Cortex
- Tools: Git, MATLAB
- Networking: Cisco Packet Tracer, 5G Protocols, Zero-Trust Security

## TECHNICAL PROJECTS

**Gastrointestinal Disease Classification using Vision Transformers**

**2025**

*Research Project*

*MaxViT, Python*

- Trained a MaxViT transformer model on 8-class KVASIR v2 dataset of 8,000+ endoscopic images for multi-disease GI classification.
- Automated disease identification, contributing to 90% accuracy and enabling early detection in clinical settings
- Attained high accuracy using data augmentation, plotted training graphs and confusion matrix.

**Portfolio Website**

**2025**

*Frontend Project*

*HTML, CSS, JavaScript*

- Created and deployed a personal portfolio website to showcase my resume, technical skills, and 5+ project demonstrations.
- Developed an interface that is fully responsive using custom CSS to make it accessible and compatible across gadgets with ease.
- Live: Karthik's Portfolio

**Travel Expense Tracker Web App-"Wayfare Wallet"**

**2025**

*Frontend Project*

*HTML, CSS, JavaScript*

- Designed and built an interactive web application to log, categorize, and analyze personal travel expenses upto a limit of 5 members at a time.
- Released the entire project to GitHub showing quality of code, ability to follow modularity and with practical frontend knowledge.
- GitHub: Wayfare Wallet

**AI-based Acoustic Wave Monitoring System**

**2025**

*Applied Patent*

*Edge Computing*

- Built a rail defect detection system processing 500,000 audio samples, achieving 90% accuracy and reducing inspection time by 40%.
- The deep learning model required the processing of more than 500,000 audio samples for real-time fault detection purposes.
- The AI-based monitoring framework received patent approval through a published document which led to a 40% enhancement of predictive maintenance efficiency.

### **Deepfake Detection Using SqueezeNet**

**2024**

*Research Project*

*Deep Learning*

- Achieved 98.21% validation accuracy on 260,000+ images with SqueezeNet, outperforming conventional models by 5% in speed and memory efficiency.
- Highlighted the suitability of lightweight neural networks like SqueezeNet for scalable, accurate classification tasks in real-world scenarios.
- Validated high model performance with extensive testing, confirming robust generalization across a diverse image set.

### **Smart India Hackathon Winner – Drone-Based Magnetic Sensing System**

**2023**

*Smart India Hackathon*

*Python, Embedded Systems, IOT*

- Enhanced: Led team to design a UAV-based sensor system with 40% higher accuracy, supporting 25% longer flight times and processing 20,000+ sensor inputs during competition.
- Extended drone flight time by 25% through power-efficient design.
- The system delivered a 65% reduction in data delay during the national final alongside the solution of 7 technical problems.

## **CERTIFICATIONS**

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### **Technical Certifications:**

- Neural Networks and Deep Learning - Coursera, Andrew Ng
- Supervised Machine Learning: Regression and Classification - Coursera, Andrew Ng
- The Complete Full-Stack Web Development Bootcamp - Udemy, Dr. Angela Yu
- C++ Programming Course – GeeksforGeeks (Certified)
- Smart Interviews Certification – Specialized training in DSA, problem-solving, and coding interviews.

## **ACHIEVEMENTS**

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### **Technical Achievements:**

- Winner of Smart India Hackathon 2023 under the Ministry of Defence, competing among 30,000+ participants.
- Published Patent for AI-based Acoustic Wave Monitoring System, improving defect detection rates by 40%.
- Construction and implementation of personal web page with display of resume and technology portfolios.